

1. Introduction

This Software Requirements Specification (SRS) document provides a detailed and structured description of the requirements for the **Taqwa Discount Shop Management System**. The purpose of this section is to present an overall understanding of the system, including its objectives, scope, and intended functionality.

The document outlines the system's primary goals, core features, and operational boundaries. It also identifies the target users of the system and establishes a common terminology to ensure clarity and consistency throughout the document. By clearly defining expectations and requirements, this SRS serves as a communication bridge between stakeholders and the development team and acts as a reference throughout the software development lifecycle.

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) document is to clearly define the functional and non-functional requirements of the **Taqwa Discount Shop Management System**. The system is designed to streamline and automate daily retail operations, including billing, inventory management, supplier handling, customer management, staff administration, and financial reporting.

This document ensures that all stakeholders—such as the shop owner, managers, staff members, and the development team—share a common and unified understanding of the system's objectives, features, and constraints. It provides a structured reference that guides system design, development, testing, and implementation.

Ultimately, the goal of this document is to support the development of a reliable, secure, and user-friendly management system that effectively meets the operational needs of a modern discount retail shop.

1.2 Scope

The **Taqwa Discount Shop Management System** is an integrated retail management solution developed to automate, optimize, and centralize daily shop operations. The system will provide a single platform through which shop owners, managers, and staff can efficiently manage key business processes, including the following:

Point of Sale (POS) & Billing

- Real-time product scanning
- Automatic discount and VAT calculation
- Invoice generation and receipt printing

Inventory Management

- Real-time stock level monitoring
- Product detail management and categorization
- Reorder level configuration and low-stock alerts

Supplier & Purchase Order Management

- Supplier information management
- Purchase order creation and tracking
- Inventory updates upon product delivery

Customer Management

- Customer profile and contact information storage
- Purchase history tracking
- Loyalty and repeat-customer program management

Financial Tracking

- Daily sales and expense recording
- Transaction categorization
- Secure audit trails for financial transparency

Staff Management

- Role-based access control
- User account and credential management
- Controlled system access based on permissions

Reporting & Analytics

- Sales performance reports
- Inventory summaries
- Financial statements and profit analysis
- Customer behavior and insight reports

The system will be implemented as a **secure desktop-based application**, with provisions for future mobile platform compatibility. It will support integration with external hardware such as barcode scanners and receipt printers, and ensure real-time data synchronization.

To safeguard sensitive business and customer data, the system will incorporate robust security mechanisms, including encrypted data storage and role-based access control.

Overall, the **Taqwa Discount Shop Management System** aims to improve operational efficiency, minimize manual errors, enhance customer service quality, and enable data-driven decision-making for shop owners and managers.

1.3 Definitions, Acronyms, and Abbreviations

This section defines the key terms, acronyms, and abbreviations used throughout this Software Requirements Specification (SRS) document. These definitions are provided to ensure clarity, consistency, and a common understanding among all stakeholders involved in the development and use of the **Taqwa Discount Shop Management System**.

Term/Acronym	Definition
POS	Point of Sale – The system used to process customer purchases, including item scanning, billing, and payment handling.
VAT	Value Added Tax-A consumption tax applied to goods and services,automatically calculated during the billing process
SKU	Stock Keeping Unit – A unique identifier assigned to each product for inventory tracking and management.
UC	Use Case – A description of a specific interaction between a user (actor) and the system to achieve a goal.
ERD	Entity Relationship Diagram – A visual representation of database entities and their relationships.
PO	Purchase Order – A formal document issued to suppliers for ordering and procuring products.
CRM	Customer Relationship Management – A system used to manage customer information, interactions, and purchase history.
Barcode Scanner	A hardware device used to scan product barcodes for quick and accurate item identification during billing.
Low Stock Alert	A system-generated notification triggered when product quantity falls below the predefined inventory threshold.

Loyalty Discount	A discount offered to eligible customers based on accumulated loyalty points or purchase history.
Invoice	A detailed bill generated for each customer transaction, listing items, quantities, prices, taxes, discounts, and total amount.
Inventory Threshold	The minimum stock level defined for a product; reaching this level triggers a restocking alert.
Owner/Admin	The primary system administrator with full access to all system modules, settings, reports, and user management features.
Manager	A system user role authorized to manage sales, inventory, customer data, and generate operational reports.
Staff	A system user role with limited access, primarily responsible for sales assistance, billing, stocking, and customer support.
Real Time Update	Immediate synchronization of data across the system, such as automatic stock reduction after a sale.
DBMS	Database Management System – Software used to store, manage, retrieve, and secure system data.
Taqwa Discount Shop	The retail business entity for which this management system is being developed

RBAC	A security mechanism that restricts system access based on assigned user roles such as Owner, Manager, and Staff.
Payment Mode	The method used to complete a transaction, such as cash, card, or mobile payment.
Expiry Date Tracking	A system feature that monitors perishable products and alerts users when items approach their expiration date.

1.4 References

The following references were consulted in the preparation of this Software Requirements Specification (SRS) document for the **Taqwa Discount Shop Management System**. These sources provide industry standards, best practices, regulatory guidelines, and technical insights relevant to retail management system development.

1. **IEEE 830-1998 Standard**
IEEE Recommended Practice for Software Requirements Specifications – Provides standardized guidelines for structuring and documenting software requirements.
2. **Retail Management Systems Case Studies (2023)**
Studies and analyses of successful POS and inventory management systems used in discount retail environments.
3. **Bangladesh VAT & Taxation Guidelines**
Official documentation outlining VAT rules, tax calculation methods, and compliance requirements for retail businesses in Bangladesh.
4. **Inventory Management Best Practices**
Industry-recognized methodologies for efficient stock control, reorder point management, and supplier coordination.
5. **User Interface Design Principles for Retail Systems**
Guidelines for developing intuitive, efficient, and user-friendly interfaces tailored to retail staff and management needs.

6. **Data Security & Privacy Compliance Documentation**

Standards and best practices for protecting customer, transaction, and business data in compliance with local data protection regulations.

7. **Hardware Integration Guides**

Technical documentation for integrating barcode scanners, receipt printers, and cash drawers with POS software.

8. **Database Design – Modern Approaches**

References covering relational database design principles, normalization techniques, and ER modeling for retail systems.

9. **Taqwa Discount Shop Business Process Documentation (2024)**

Internal analysis of existing shop workflows, operational challenges, and requirements gathered from key stakeholders.

10. **Agile Software Development Methodology**

Frameworks and practices supporting iterative development, continuous testing, and incremental deployment of the system.

2. Overall Description

This section presents an overall description of the **Taqwa Discount Shop Management System**, highlighting its main features, functional capabilities, and operational environment. It explains how the system fits into the existing retail workflow, identifies its intended users, and outlines key constraints and dependencies. This overview provides stakeholders with a clear understanding of the system's objectives, structure, and boundaries.

2.1 Product Perspective

The **Taqwa Discount Shop Management System** is an integrated retail management solution designed to streamline and automate daily operations for small to medium-sized discount retail stores. The system replaces manual and semi-automated processes with a centralized, digital platform that enhances accuracy, efficiency, and operational control.

The system automates core retail activities such as **point-of-sale (POS) billing, inventory management, supplier and purchase order handling, customer relationship management, staff administration, and financial tracking**. It is designed to operate primarily as a desktop-based application, with support for future mobile and tablet compatibility.

To enhance operational flexibility, the system supports optional integration with external hardware devices, including **barcode scanners, receipt printers, and cash drawers**.

In addition, the system can interact with external services such as:

- **Payment Gateways** for card and mobile-based transactions
- **Supplier Portals** for automated purchase order communication
- **Local VAT/TAX APIs** for real-time tax calculation and compliance
- **SMS and Email Services** for customer notifications, promotions, and system alerts

The system architecture is modular in nature, consisting of the following core modules:

- Sales & Billing
- Inventory Management
- Customer Management
- Supplier & Purchase Order Management
- Financial Tracking
- Staff Management
- Reporting & Analytics

Product Perspective Diagram

The following diagram illustrates the high-level interaction between users, system modules, and external services.

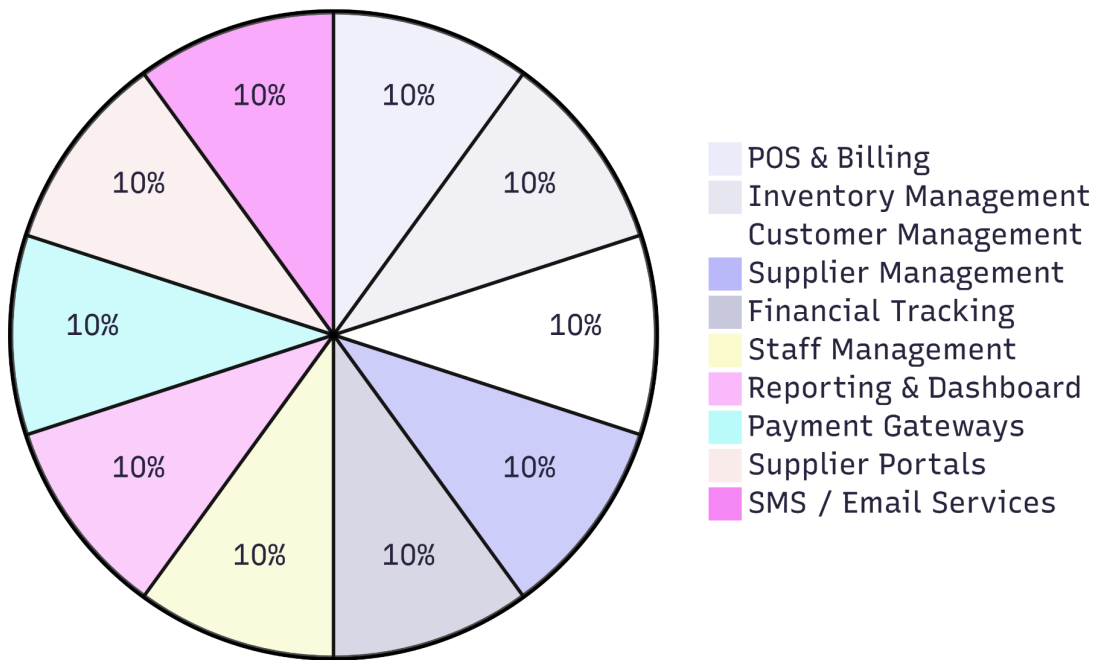


Figure: Product Perspective diagram

2.2 Product Functions

The **Taqwa Discount Shop Management System** provides a comprehensive set of functions to support efficient, accurate, and secure shop operations. The system begins with a **secure authentication mechanism**, ensuring that users can access features based on their assigned roles (Owner, Manager, or Staff).

Once authenticated, the system supports the following core functions:

Sales Processing & Invoice Generation

- Barcode-based item scanning
- Automatic discount and VAT calculation
- Invoice and receipt generation
- Support for multiple payment modes (cash, card, mobile payment)

Real-Time Inventory Management

- Add, update, and remove product records
- Set inventory thresholds and receive low-stock alerts
- Monitor stock movement in real time
- Track expiry dates for perishable products

Customer Relationship Management

- Create and maintain customer profiles
- Track customer purchase history
- Manage loyalty points and discounts
- Send promotional messages via SMS or email

Supplier & Purchase Order Management

- Maintain supplier records and contact information
- Generate and track purchase orders
- Update inventory upon delivery confirmation

Financial Tracking & Accounting Support

- Record daily sales and operational expenses
- Categorize expenses (rent, utilities, salaries, etc.)
- Generate profit and loss summaries

Staff Management

- Register staff accounts and assign roles

- Manage login credentials and permissions
- Monitor staff activity through system logs

Reporting & Analytics

- Sales reports (daily, weekly, monthly)
- Inventory summaries and stock valuation reports
- Financial performance reports
- Customer purchase trends and loyalty analysis

System Notifications & Alerts

- Low-stock notifications
- Expiry date reminders
- Sales and performance milestone alerts

Product Functions Block Diagram

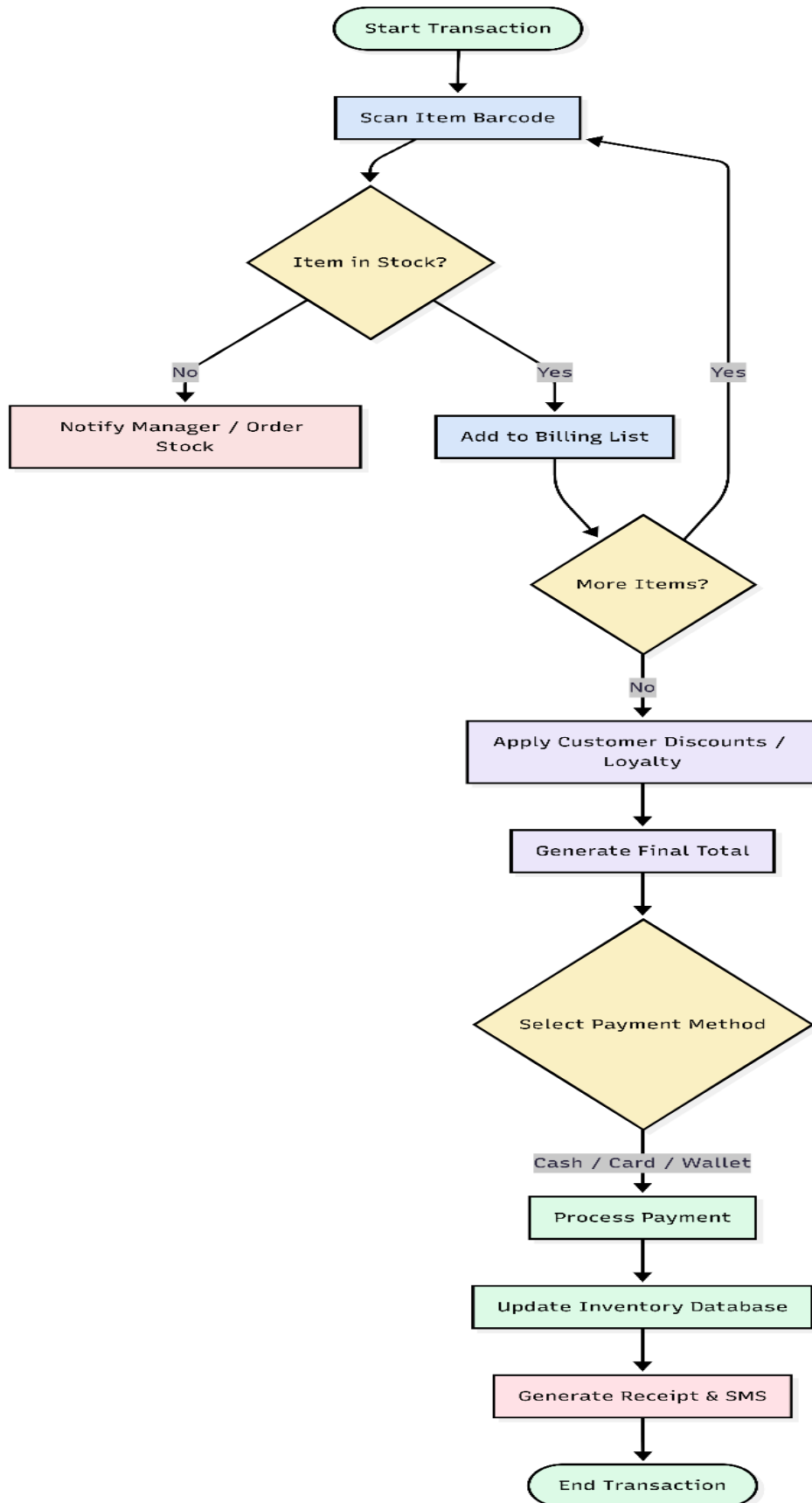


Figure: Product functions block diagram

The system is designed to be **user-friendly, fast, and reliable**, allowing shop staff to perform daily tasks with minimal training. Real-time data synchronization ensures that sales, inventory, and financial data remain consistent and up to date across all modules, supporting accurate reporting and informed decision-making.

2.3 User Characteristics

The **Taqwa Discount Shop Management System** serves multiple user groups, each with distinct responsibilities, access levels, and interface needs:

User Group	Description	Key Needs
Owner (Admin)	Full administrative access; oversees business, finances, suppliers, staff, and system configuration	Comprehensive reporting, audit trails, full system control, high-level insights, secure access
Manager	Daily operational users handling POS, inventory, customer service, and basic financial entries	Fast, intuitive interface, real-time updates, quick access to sales, stock, and customer data
Staff	Assists in billing, stocking, and customer service	Simple, guided interface, minimal training, access mainly to POS and product lookup
Supplier (External)	Interacts indirectly via POs and delivery updates	Timely notifications, delivery confirmations, integrated communication (SMS/email)
Customer (Indirect)	Benefits from faster checkout, accurate pricing, loyalty rewards	Accurate billing, loyalty tracking, receipt generation, secure data storage

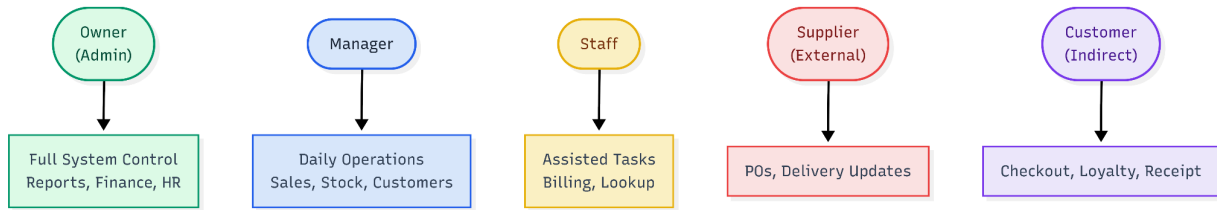


Figure: User Characteristics block diagram

2.4 Constraints

The system will operate under several **technical, operational, and regulatory constraints**:

1. **Performance:**

- Handle high-volume transactions during peak hours
- Billing < 3 seconds, Inventory lookup < 2 seconds

2. **Hardware Dependencies:**

- Barcode scanners, receipt printers, cash drawers
- Must support common hardware models

3. **Connectivity:**

- Offline sales logging available
- Full functionality (real-time inventory, payments, reports) requires stable internet

4. **Regulatory Compliance:**

- Adherence to Bangladesh VAT/tax rules, financial standards, data protection laws

5. **User Skill Level:**

- Staff with minimal technical knowledge
- Intuitive, icon-driven interface

6. Scalability:

- Initially single-shop
- Should allow multi-shop expansion

7. Integration Dependencies:

- Third-party services: payment gateways, SMS, supplier portals
- Downtime or changes may impact functionality

8. Budget & Resource Limits:

- Must align with shop budget
- Avoid overly complex or expensive infrastructure

2.5 Assumptions & Dependencies

Assumptions

- Staff trained in standardized procedures (sales, returns, stock entry)
- Products have correctly printed barcodes
- Electricity & internet generally stable
- Suppliers deliver as per POs with accurate documentation
- Owner/Manager reviews reports and acts on alerts
- Customers provide valid contact information
- System initially deployed in a single physical location

Dependencies

- **Hardware:** Barcode scanners, printers, cash drawers

- **External Services:** Payment gateway APIs, SMS/email notifications, supplier portals
- **Regulatory:** VAT policies, retail regulations, customer data protection
- **Infrastructure:** Stable electricity, internet, secure database hosting
- **Human Resources:** Trained staff, managerial oversight

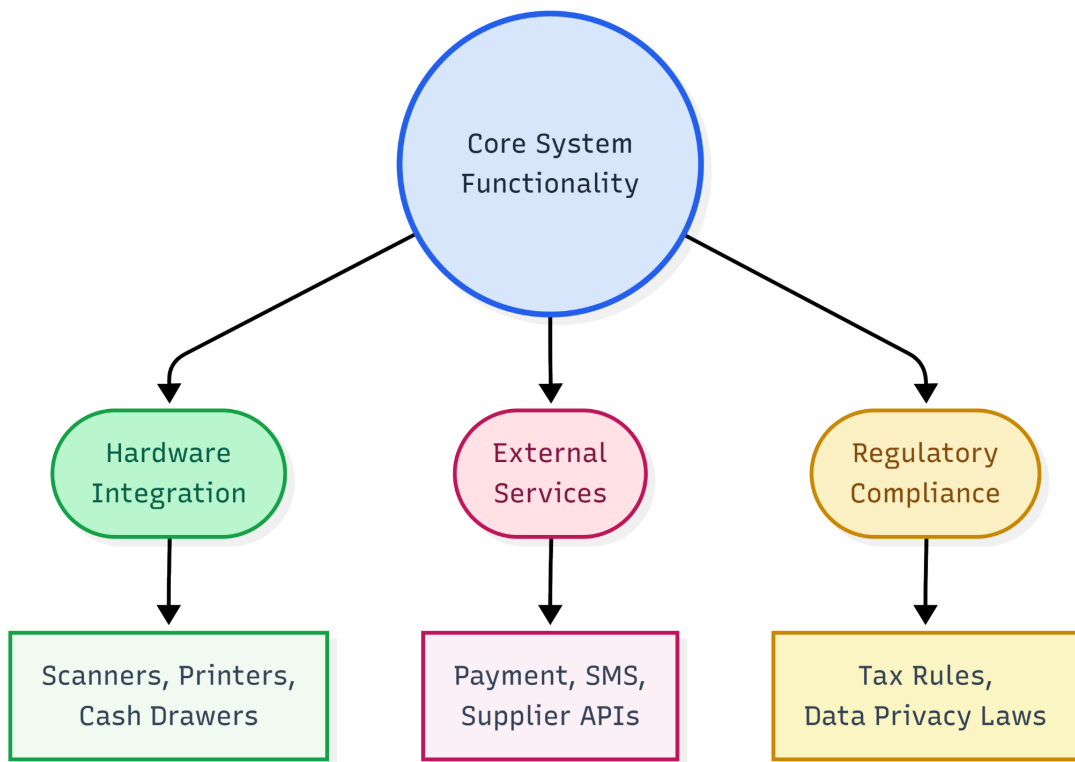


Figure: Assumptions & Dependencies

3. User Interface and Specific Requirements

3.1 User Interface and Functional Requirements

3.1.1 Customer Purchase and Billing

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Req ID	Requirement Description	Priority	Acceptance Criteria
FR-1.1	The system shall allow managers to scan product barcodes to add items to the transaction	High	Barcode scanning completes within 1 second and displays product details
FR-1.2	The system shall automatically calculate and apply VAT to the total amount	High	VAT calculation is accurate to 2 decimal places and displayed in real-time
FR-1.3	The system shall support multiple discount types (loyalty and seasonal) based on customer eligibility	Medium	Discounts are applied automatically when customer is identified; manual override available
FR-1.4	The system shall display running total, applicable discounts, and final payable amount during billing	High	All amounts update in real-time as items are scanned
FR-1.5	The system shall support both cash and card payment modes	High	Payment mode is selectable and recorded in transaction log

FR-1 .6	The system shall generate and print invoices immediately after payment confirmation	High	Invoice generation and printing completes within 5 seconds
FR-1 .7	The system shall automatically deduct sold quantities from inventory in real-time	Critical	Stock levels update immediately upon transaction completion
FR-1 .8	The system shall validate sufficient stock availability before completing the transaction	Critical	System prevents sale if stock is insufficient and alerts user
FR-1 .9	The system shall record transaction timestamp and user details for audit trail	Medium	All transactions are timestamped and linked to logged-in manager
FR-1 .10	The system shall allow transaction cancellation before payment confirmation	Medium	Cancelled transactions restore stock and do not generate invoice

3.1.2 Inventory Management

[UI Screenshot Placeholder]

Req ID	Requirement Description	Priority	Acceptance Criteria
FR-2.1	The system shall allow managers to add new products with mandatory fields (ID, name, category, quantity, threshold)	High	New products are saved with auto-generated product ID

FR-2.2	The system shall enable managers to edit existing product information except product ID	High	Changes are validated and saved within 2 seconds
FR-2.3	The system shall prevent deletion of products linked to pending transactions	Critical	System displays error message and blocks deletion attempt
FR-2.4	The system shall display real-time stock levels for all products	High	Stock quantities update immediately after any transaction
FR-2.5	The system shall generate low-stock alerts when quantity falls below threshold	High	Alert is triggered within 5 seconds of stock falling below threshold
FR-2.6	The system shall support product categorization and brand information	Medium	Products can be filtered and searched by category and brand
FR-2.7	The system shall track product expiry dates for perishable items	Medium	System highlights products nearing expiry (within 30 days)
FR-2.8	The system shall provide search functionality by product name, ID, or barcode	High	Search results display within 1 second
FR-2.9	The system shall validate barcode uniqueness during product creation	High	System prevents duplicate barcodes and prompts update option

FR-2.10	The system shall display inventory data to billing module in real-time	Critical	Billing system always accesses current stock levels
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3.1.3 Supplier and Purchase Management

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Req ID	Requirement Description	Priority	Acceptance Criteria
FR-3.1	The system shall allow owner to register new suppliers with contact information	High	Supplier records are saved with auto-generated supplier ID
FR-3.2	The system shall enable owner to create purchase orders with product list and quantities	High	Purchase orders are created with unique order ID and pending status
FR-3.3	The system shall restrict purchase order creation and approval to owner only	Critical	Managers can only view purchase orders, not create or modify
FR-3.4	The system shall track purchase order status (Pending/Delivered)	High	Status is updateable only by owner and visible to managers
FR-3.5	The system shall update inventory automatically upon delivery confirmation	Critical	Stock quantities increase immediately when order status changes to Delivered

FR-3.6	The system shall record expected and actual delivery dates	Medium	Delivery dates are tracked and displayed in order history
FR-3.7	The system shall allow owner to edit or delete supplier information	Medium	Changes to supplier data are validated and saved securely
FR-3.8	The system shall generate purchase orders based on low-stock alerts	Medium	System suggests products below threshold for new purchase orders
FR-3.9	The system shall maintain complete purchase history with supplier details	Medium	All purchase orders are archived and searchable by date/supplier
FR-3.10	The system shall validate supplier contact information uniqueness	Low	System alerts if duplicate contact number is entered

3.1.4 Customer Management

[UI Screenshot Placeholder]

Req ID	Requirement Description	Priority	Acceptance Criteria
FR-4.1	The system shall allow managers to register new customers with mandatory fields (name, phone)	High	Customer records are created with auto-generated customer ID

FR-4.2	The system shall enforce unique phone numbers for each customer	Critical	System prevents duplicate phone numbers and suggests update option
FR-4.3	The system shall enable managers to edit customer information except customer ID	Medium	Customer updates are validated and saved within 2 seconds
FR-4.4	The system shall track customer loyalty eligibility based on purchase history	Medium	Loyalty status is automatically calculated and displayed
FR-4.5	The system shall record customer join date automatically upon registration	Low	Join date is auto-filled with current date during creation
FR-4.6	The system shall provide customer search by name, phone, or customer ID	High	Search results display within 1 second with matching records
FR-4.7	The system shall link customer profiles to transaction history	High	All customer purchases are viewable in their profile
FR-4.8	The system shall allow staff to assist in collecting customer data during registration	Medium	Staff can access customer registration interface with manager supervision
FR-4.9	The system shall enable owner to view customer analytics and loyalty trends	Medium	Owner dashboard displays customer statistics and reports

FR-4.10	The system shall support optional address field for customer profiles	Low	Address can be added or left blank without validation errors
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3.1.5 Expense and Income Tracking

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Req ID	Requirement Description	Priority	Acceptance Criteria
FR-5.1	The system shall allow managers to log expense transactions with category and amount	High	Expense entries are saved with auto-generated entry ID and timestamp
FR-5.2	The system shall automatically record sales income from completed transactions	Critical	Sales are logged immediately upon invoice generation
FR-5.3	The system shall categorize transactions as either "Sale" or "Expense"	High	Transaction type is mandatory and validated during entry
FR-5.4	The system shall capture the manager's identity who logged each entry	Medium	Entry records include username from current session
FR-5.5	The system shall restrict modification/deletion of financial records to owner only	Critical	Managers can only create entries; owner can approve and modify

FR-5.6	The system shall support expense categorization (Utilities, Rent, Supplies, etc.)	Medium	Categories are selectable from predefined list or custom entry
FR-5.7	The system shall validate that expense amounts are positive numbers	High	System rejects zero or negative amounts with error message
FR-5.8	The system shall auto-fill transaction date with current date/time	Medium	Date can be auto-filled or manually adjusted by user
FR-5.9	The system shall provide daily income/expense summary for managers	Medium	Summary displays total sales and expenses for selected date
FR-5.10	The system shall maintain complete audit trail of all financial transactions	High	All entries are timestamped and non-deletable by managers

3.1.6 Staff Management

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Req ID	Requirement Description	Priority	Acceptance Criteria
FR-6.1	The system shall allow owner to register new staff with role assignment (Manager/Staff)	Critical	Staff records are created with auto-generated staff ID

FR-6.2	The system shall enforce unique usernames for all staff accounts	Critical	System prevents duplicate usernames and prompts alternative
FR-6.3	The system shall store staff passwords using secure encryption	Critical	Passwords are hashed and never displayed in plain text
FR-6.4	The system shall restrict all staff management functions to owner only	Critical	Only owner can access staff module; others receive access denied error
FR-6.5	The system shall record staff join date automatically during registration	Low	Join date is auto-filled with current date upon creation
FR-6.6	The system shall enable owner to update staff roles and credentials	High	Role changes require confirmation before saving
FR-6.7	The system shall allow owner to deactivate or remove staff accounts	High	Removed staff cannot log in; their data remains in archive
FR-6.8	The system shall assign access privileges based on staff role	Critical	Managers have full operational access; Staff have limited access
FR-6.9	The system shall maintain staff activity logs for accountability	Medium	System tracks login times and actions performed by each staff member
FR-6.10	The system shall require confirmation before removing staff accounts	Medium	Owner must confirm deletion with Yes/No dialog

3.1.7 Invoice and Report Generation

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Req ID	Requirement Description	Priority	Acceptance Criteria
FR-7.1	The system shall generate invoices automatically upon transaction completion	Critical	Invoices include all transaction details, timestamps, and unique invoice ID
FR-7.2	The system shall allow managers to generate sales and inventory reports	High	Reports are generated within 5 seconds for selected date range
FR-7.3	The system shall enable owner to access all report types (Sales, Expense, Inventory, Analytics)	High	Owner dashboard provides access to comprehensive reporting
FR-7.4	The system shall allow staff to view their daily sales summaries	Medium	Staff can access reports limited to their own transactions
FR-7.5	The system shall support report filtering by date range	High	Users can select start and end dates for custom reports
FR-7.6	The system shall provide report export in multiple formats (PDF, Excel)	Medium	Reports are downloadable in selected format without data loss
FR-7.7	The system shall generate reports with unique report ID for tracking	Low	Each generated report has auto-generated identifier

FR-7.8	The system shall display "No data available" message when report is empty	Low	System gracefully handles empty result sets
FR-7.9	The system shall log report generation with user identity and timestamp	Medium	Audit trail captures who generated which report and when
FR-7.10	The system shall restrict report types based on user role permissions	Critical	Staff cannot access financial or expense reports

3.1.8 Authentication and Authorization

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Req ID	Requirement Description	Priority	Acceptance Criteria
FR-8.1	The system shall require username and password for all users to log in	Critical	Users cannot access any functionality without authentication
FR-8.2	The system shall validate credentials against stored staff records	Critical	Invalid credentials display error message and deny access
FR-8.3	The system shall implement role-based access control (Owner, Manager, Staff)	Critical	Each role has predefined permissions enforced throughout the system

FR-8.4	The system shall maintain active session for logged-in users	High	Session remains active until user logs out or timeout occurs
FR-8.5	The system shall lock accounts after 5 consecutive failed login attempts	High	Locked accounts require owner intervention to unlock
FR-8.6	The system shall log all login attempts with timestamp and outcome	Medium	Audit trail includes successful and failed login records
FR-8.7	The system shall automatically log out users after 30 minutes of inactivity	Medium	Inactive sessions are terminated to prevent unauthorized access
FR-8.8	The system shall allow users to change their own passwords	Medium	Password changes require current password verification
FR-8.9	The system shall enforce password complexity requirements (minimum 8 characters)	High	Weak passwords are rejected during creation or change
FR-8.10	The system shall display current logged-in user's name and role in the interface	Low	User identity is visible in header or profile section

3.1.9 Data Backup and Recovery

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Req ID	Requirement Description	Priority	Acceptance Criteria
FR-9.1	The system shall perform automatic daily backups of the entire database	Critical	Backups are created every 24 hours without user intervention
FR-9.2	The system shall allow owner to initiate manual backup on demand	High	Manual backup completes within 2 minutes with success confirmation
FR-9.3	The system shall store backups in a secure, separate location from primary database	Critical	Backup files are encrypted and stored on different physical/virtual storage
FR-9.4	The system shall maintain backup history for at least 30 days	High	System keeps rolling 30-day backup archive
FR-9.5	The system shall enable owner to restore database from any available backup	Critical	Restore process includes confirmation and completes within 5 minutes
FR-9.6	The system shall validate backup integrity after each backup operation	High	Corrupted backups are detected and user is alerted
FR-9.7	The system shall log all backup and restore operations with timestamp	Medium	Audit trail includes backup/restore status and user identity
FR-9.8	The system shall notify owner of backup failures via system alert	High	Failed backups trigger immediate notification

FR-9.9	The system shall support incremental backups to optimize storage	Medium	Only changed data is backed up after initial full backup
FR-9.10	The system shall provide backup scheduling options (daily, weekly, custom)	Low	Owner can configure backup frequency in settings

3.1.10 System Administration

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Req ID	Requirement Description	Priority	Acceptance Criteria
FR-10.1	The system shall allow owner to configure system-wide settings (VAT rate, shop name, etc.)	High	Settings are saved and applied across all modules immediately
FR-10.2	The system shall enable owner to view system activity logs and audit trails	High	Logs display all user actions with timestamps and details
FR-10.3	The system shall allow owner to configure low-stock alert thresholds globally	Medium	Global threshold can be set or overridden per product
FR-10.4	The system shall provide database maintenance tools (optimize, repair)	Medium	Maintenance operations complete without data loss

FR-10.5	The system shall enable owner to configure automatic logout timeout duration	Low	Timeout value is adjustable from 10 to 60 minutes
FR-10.6	The system shall allow owner to enable/disable specific modules or features	Medium	Disabled features are hidden from all users except owner
FR-10.7	The system shall provide system health monitoring (disk space, performance)	Medium	Dashboard displays current system status and resource usage
FR-10.8	The system shall allow owner to configure invoice templates and formats	Low	Custom templates are applied to all generated invoices
FR-10.9	The system shall enable owner to manage discount rules and eligibility criteria	Medium	Discount rules are configurable and applied automatically at billing
FR-10.10	The system shall maintain system configuration change history	Low	All configuration changes are logged with user and timestamp

3.2 External Interface Requirements

3.2.1 User Interfaces

Req ID	Requirement Description	Priority	Acceptance Criteria
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UI-1	The system shall provide a responsive web-based interface accessible via modern browsers	High	Interface works on Chrome, Firefox, Edge, Safari (latest 2 versions)
UI-2	The system shall display role-specific dashboards upon successful login	High	Dashboard content varies based on user role (Owner/Manager/Staff)
UI-3	The system shall use intuitive navigation with clearly labeled menus and buttons	Medium	All functions are accessible within 3 clicks from dashboard
UI-4	The system shall provide consistent color scheme and layout across all modules	Low	UI follows a unified design pattern throughout
UI-5	The system shall display error messages in clear, non-technical language	Medium	Error messages guide users on how to resolve issues
UI-6	The system shall support keyboard shortcuts for frequent operations	Low	Common actions (billing, search) have keyboard shortcuts
UI-7	The system shall provide confirmation dialogs for critical actions (delete, payment)	High	Users must confirm before irreversible operations execute

3.2.2 Hardware Interfaces

Req ID	Requirement Description	Priority	Acceptance Criteria
HW-1	The system shall interface with barcode scanners via USB or wireless connection	Critical	Scanned barcodes are captured and processed instantly
HW-2	The system shall support thermal and inkjet receipt printers	High	Invoices print correctly on configured printer models
HW-3	The system shall interface with cash drawers triggered by printer signal	Medium	Cash drawer opens automatically after successful payment
HW-4	The system shall support card payment terminals via standard POS integration	Medium	Card transactions are processed and recorded in the system
HW-5	The system shall function on standard PC hardware with minimum specifications	High	System runs smoothly on minimum 4GB RAM, dual-core processor

3.2.3 Software Interfaces

Req ID	Requirement Description	Priority	Acceptance Criteria
SW-1	The system shall use a relational database management system (MySQL/PostgreSQL)	Critical	All data is stored and retrieved using RDBMS

SW-2	The system shall integrate with payment gateway APIs for card transactions	Medium	Card payments are processed securely through API
SW-3	The system shall support PDF generation libraries for invoice and report export	High	PDFs are generated accurately with proper formatting
SW-4	The system shall use secure communication protocols (HTTPS/TLS) for data transmission	Critical	All network communication is encrypted
SW-5	The system shall be compatible with Windows 10/11 and Linux operating systems	High	Application runs without errors on specified OS versions

3.2.4 Communication Interfaces

Req ID	Requirement Description	Priority	Acceptance Criteria
COM-1	The system shall communicate with database server via TCP/IP network protocol	Critical	Database connection is established and maintained reliably
COM-2	The system shall use HTTP/HTTPS for web interface communication	Critical	Web requests are processed securely over encrypted connection

COM-3	The system shall support email notifications for critical alerts (backup failures, system errors)	Medium	Email alerts are sent to owner's configured email address
COM-4	The system shall maintain persistent connection with POS peripherals	High	Barcode scanners and printers remain connected during operation
COM-5	The system shall support LAN-based multi-terminal deployment	Medium	Multiple workstations can access the system simultaneously

3.3 Non-Functional Requirements

3.3.1 Performance Requirements

Req ID	Requirement Description	Priority	Acceptance Criteria
PERF-1	The system shall respond to barcode scans within 1 second	Critical	Product details display immediately after scan
PERF-2	The system shall complete database operations (save, update) within 2 seconds	High	User receives confirmation within 2 seconds
PERF-3	The system shall generate reports within 5 seconds for up to 10,000 records	High	Reports display or download without timeout

PERF-4	The system shall support at least 5 concurrent users without performance degradation	High	Response time remains consistent with multiple active sessions
PERF-5	The system shall print invoices within 5 seconds of payment confirmation	High	Printing completes before next transaction begins
PERF-6	The system shall maintain database query performance with up to 100,000 transactions	Medium	Search and reporting remain fast as data grows

3.3.2 Safety Requirements

Req ID	Requirement Description	Priority	Acceptance Criteria
SAFE-1	The system shall prevent data corruption during unexpected shutdowns	Critical	Database transactions are atomic and recoverable
SAFE-2	The system shall validate all user inputs to prevent invalid data entry	High	Invalid inputs are rejected with clear error messages
SAFE-3	The system shall prevent inventory overselling by validating stock before sale	Critical	Transactions are blocked if stock is insufficient
SAFE-4	The system shall maintain referential integrity across all database tables	Critical	Orphaned records are prevented through foreign key constraints

SAFE-5	The system shall provide rollback mechanism for failed transactions	High	Failed operations do not leave database in inconsistent state
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3.3.3 Security Requirements

Req ID	Requirement Description	Priority	Acceptance Criteria
SEC-1	The system shall encrypt all passwords using industry-standard hashing (bcrypt, SHA-256)	Critical	Passwords are never stored or transmitted in plain text
SEC-2	The system shall implement role-based access control enforced at application and database level	Critical	Users can only access authorized functions and data
SEC-3	The system shall log all sensitive operations (login, financial transactions, data modifications)	High	Audit trail is complete and tamper-proof
SEC-4	The system shall protect against SQL injection through parameterized queries	Critical	All database queries use prepared statements
SEC-5	The system shall implement session management with secure tokens	High	Sessions are protected against hijacking and fixation
SEC-6	The system shall enforce HTTPS for all web communications	High	Data transmission is encrypted end-to-end

SEC-7	The system shall automatically expire inactive sessions after configurable timeout	Medium	Unattended terminals are protected from unauthorized access
SEC-8	The system shall restrict backup access to owner only with authentication	Critical	Backup files are protected from unauthorized access

3.3.4 Software Quality Attributes

Req ID	Requirement Description	Priority	Acceptance Criteria
QA-1	Reliability: The system shall maintain 99% uptime during business hours	High	System is available and functional 99% of operating time
QA-2	Maintainability: The system shall use modular architecture for easy updates	Medium	Individual modules can be updated without affecting others
QA-3	Usability: The system shall be learnable by new users within 2 hours of training	Medium	Users can perform basic operations after brief training
QA-4	Portability: The system shall be deployable on different operating systems without modification	Medium	Application works on Windows and Linux without code changes

QA-5	Accuracy: The system shall calculate financial amounts with precision up to 2 decimal places	Critical	All monetary calculations are accurate without rounding errors
QA-6	Testability: The system shall provide logging mechanisms for debugging and testing	Medium	Logs capture sufficient detail for troubleshooting

3.3.5 Business Rules

Req ID	Business Rule Description	Priority	Enforcement
BR-1	Discounts can only be applied to eligible customers based on loyalty status	High	System validates eligibility before applying discount
BR-2	Inventory must be updated immediately upon transaction completion	Critical	Stock deduction is part of atomic transaction
BR-3	Only owner can approve purchase orders and manage suppliers	Critical	Role-based access control enforces this restriction
BR-4	Products linked to pending transactions cannot be deleted	Critical	System blocks deletion and displays error message
BR-5	Each phone number must be unique in customer database	High	Database constraint enforces uniqueness

BR-6	Financial records can only be modified by owner after initial submission	Critical	Managers have read-only access to submitted records
BR-7	VAT rate is fixed and applied uniformly to all transactions	High	VAT percentage is system-wide configuration
BR-8	Staff usernames must be unique across the system	Critical	Database constraint prevents duplicate usernames
BR-9	Low-stock alerts are triggered when quantity falls below threshold	High	Alert generation is automatic and immediate
BR-10	Invoices must include all transaction details and unique invoice ID	Critical	Invoice template includes all mandatory fields

3.4 Design Constraints

Constraint ID	Constraint Description	Rationale
DC-1	The system must be developed using a web-based architecture	Enables multi-terminal access and easier maintenance
DC-2	The system must use a relational database (MySQL/PostgreSQL)	Ensures data integrity and supports complex queries

DC-3	The system must comply with local VAT calculation and financial reporting regulations	Legal compliance requirement
DC-4	The system must support barcode standard formats (EAN-13, UPC-A)	Compatibility with common retail products
DC-5	The system must be developed within 6-month timeline	Project schedule constraint
DC-6	The system must function on existing shop hardware (no major upgrades required)	Budget constraint
DC-7	The system must support at least 3 years of transaction history	Business record-keeping requirement

3.5 Software System Attributes

Reliability

- System shall have Mean Time Between Failures (MTBF) of at least 720 hours
- System shall recover from crashes within 5 minutes without data loss
- Database transactions shall follow ACID properties (Atomicity, Consistency, Isolation, Durability)

Availability

- System shall be operational during all business hours (typically 8 AM - 10 PM)
- Planned maintenance shall occur outside business hours
- System shall provide graceful degradation if external services fail

Maintainability

- Code shall follow consistent coding standards and be well-documented
- System shall provide detailed error logs for troubleshooting

- Database schema shall be documented with entity relationships

Portability

- System shall be deployable on Windows Server, Linux Server, or cloud platforms
- System shall use standard technologies to avoid vendor lock-in
- Configuration files shall be separate from application code for easy deployment

3.6 Security

Access Control

- Multi-level role-based access (Owner, Manager, Staff) with principle of least privilege
- Strong password policy enforcement (minimum 8 characters, complexity requirements)
- Account lockout after multiple failed login attempts

Data Protection

- All sensitive data (passwords, financial records) encrypted at rest and in transit
- Regular security audits and vulnerability assessments
- Secure backup encryption with separate encryption keys

Audit and Compliance

- Complete audit trail of all financial transactions
- Tamper-proof logging of security-relevant events
- Compliance with data protection regulations (if applicable in jurisdiction)

Network Security

- HTTPS/TLS for all network communications
- SQL injection prevention through parameterized queries
- Cross-Site Scripting (XSS) and Cross-Site Request Forgery (CSRF) protection

3.7 Usability

Learnability

- Intuitive user interface with minimal learning curve

- Context-sensitive help and tooltips for complex operations
- User manual and video tutorials available

Efficiency

- Keyboard shortcuts for frequent operations (billing, search)
- Auto-complete and predictive search for product and customer lookup
- Streamlined workflows with minimal clicks required

Error Prevention and Recovery

- Input validation with immediate feedback
- Confirmation dialogs for critical actions
- Clear error messages with guidance on resolution
- Undo functionality where applicable

Accessibility

- Readable fonts and adequate color contrast
- Support for different screen sizes and resolutions
- Keyboard navigation support for all functions

3.8 Scalability

Data Scalability

- Database design shall support at least 100,000 products and 1 million transactions
- Indexing strategy to maintain query performance as data grows
- Data archiving mechanism for old transactions to manage database size

User Scalability

- System shall support up to 10 concurrent users without performance degradation
- Load balancing capability for future expansion
- Session management optimized for multiple simultaneous connections

Functional Scalability

- Modular architecture allowing addition of new features without major redesign
- API-ready design for potential third-party integrations
- Configurable system parameters to adapt to business growth

Infrastructure Scalability

- Ability to migrate from single-server to distributed architecture if needed
- Cloud deployment support for elastic scaling
- Horizontal scaling capability for database and application layers